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NCO/NITRD

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Submitted electronically via email to ostp-ai-rfi@nitrd.gov

Subject: AI Action Plan

Overview

AFL Telecommunications LLC (“AFL”) submits these comments in response to the Request for Information issued by the Networking and Information Technology Research and Development (“NITRD”) National Coordination Office (“NCO”) and National Science Foundation seeking input on the development of an Artificial Intelligence (“AI”) Action Plan. AFL believes that a key component of sustaining and enhancing America’s “AI dominance” will be ensuring that we have adequate infrastructure in place to support AI development and deployment.

Who is AFL?

AFL, headquartered in Spartanburg, South Carolina, is a leading manufacturer of optical cables and connectivity solutions. Founded 40 years ago to support the U.S. Electric Grid with optical solutions, AFL now employs thousands of American workers nationwide. The company provides essential solutions for the communication infrastructures within the U.S. Electric Grid, delivers broadband connectivity to American households, and enables the hyperscale model necessary to support Generative AI Data Centers.

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Priorities for Data Center Operators

AFL has extensive experience working with diverse data center operators, from enterprise to hyperscale. Our partnerships span a wide range of industries, focusing on delivering scalable, high-performance fiber connectivity solutions. In our experiences, we have identified the following priorities of data centers:

- **Scalability:** Flexible infrastructure to accommodate growth without major overhauls.
- **Performance:** Low-latency, high-bandwidth connectivity for demanding applications.
- **Density:** Maximizing available space for IT equipment.
- **Reliability:** Ensuring uninterrupted operations for mission-critical services.
- **Cost-Efficiency:** Optimizing total cost of ownership (TCO) through efficient designs.

Importance of Fiber Optic Technology

Optical fiber technology plays a pivotal role in addressing the critical needs of AI-driven data centers. Its unique attributes make it indispensable for ensuring the high performance and scalability required by modern AI applications. The U.S. optical fiber and cable manufacturing industry is the modern-day equivalent of the U.S. steel industry and serves as the backbone of a digitized world and enabling Generative AI Data Centers. Protecting and fostering this industry through investment and responsible trade policies should be considered a matter of national security.

Single-mode optical fiber supports speeds ranging from 10Gbps to 800Gbps and beyond, providing the necessary capacity to handle large-scale data transfers and intensive

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computational workloads. This high bandwidth is essential for enabling the seamless operation of AI systems that rely on vast amounts of data. Optical fiber offers low latency which is vital for AI applications that require immediate processing and response times, such as autonomous vehicles and real-time analytics.

In terms of scalability, optical fiber excels with its high fiber count cables and modular systems. These features allow data centers to expand their capacity seamlessly, accommodating growth without the need for major infrastructure overhauls. This flexibility is critical as AI workloads continue to increase and evolve.

Optical fiber's immunity to electromagnetic interference ensures consistent performance, which is crucial for maintaining uninterrupted operations in mission-critical environments. This reliability is essential for data centers that support AI applications, where any downtime can have significant consequences.

Recommendations for Priority Actions

- **Streamline Permitting and Rights-of-Way:** Complex permitting processes and inconsistent right-of-way regulations often delay fiber deployments. The government can standardize and simplify permitting procedures across jurisdictions, accelerating deployment timelines and reducing administrative burdens.
- **Encourage Integrated Infrastructure Planning:** Fiber deployments underpin not only data center connectivity but also grid modernization efforts such as smart grids, microgrids, renewable energy management, and electric vehicle integration. Encouraging coordinated planning between manufacturers, utilities, telecommunications providers, and data center operators can optimize

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infrastructure investments, reduce redundant construction costs, and accelerate deployment timelines.

- **Promote Public-Private Partnerships:** Government-facilitated collaboration among private-sector stakeholders—including hardware suppliers such as AFL, telecom providers, utilities, AI research institutions, and data center operators—can align interests and pool resources. Initiatives such as the NSF’s National AI Research Resource (NAIRR) pilot demonstrate how public-private cooperation can effectively support innovation infrastructure.
- **Expand Funding and Incentives:** While existing programs such as the Infrastructure Investment and Jobs Act (IIJA), Connect America Fund (CAF), and Rural Digital Opportunity Fund (RDOF) have allocated significant resources for broadband expansion, additional targeted funding or tax incentives specifically aimed at fiber deployment for data centers supporting AI, smart grids, and energy transformation could further encourage private investment.
- **Invest in Workforce Development:** A shortage of skilled technicians for fiber installation and maintenance remains a critical bottleneck. Government-supported training programs, apprenticeships, and education initiatives could expand the skilled workforce necessary to meet growing infrastructure demands.
- **Supply Chain Strategy:** The rapid growth in demand for fiber infrastructure necessitates strategic planning around fiber supply availability and capacity expansion. Ensuring adequate fiber supply to meet current and future demand requires substantial investment in manufacturing capacity, workforce development, and supply chain resilience. However, the importation of low-cost, low-quality optical fiber and cable from countries like China and India threaten the integrity of

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the U.S. fiber market. Trade remedies should be imposed to support a trade policy that protects U.S.-based optical fiber and cable manufacturers and rewards reputable foreign suppliers such as those located in Japan and Europe while putting in safeguards against the irrational and aggressive behavior of manufacturers based in China and India, as well as their foreign subsidiaries.

Respectfully Submitted,

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